

AMIS1013 SYSTEMS ANALYSIS AND DESIGN
ACADEMIC YEAR 2024/25 SESSION 202405

MIDTERM TEST

Name : _____
Programme : _____
Tutorial Group : _____



Instruction to candidates:

Total marks: 40 Marks; Test duration: 40 minutes

Answer **ALL** questions.

Scenario:

Mr. Yap, the owner of a fast food restaurant, is considering developing a self-service food ordering system that will allow customers to buy food through self-service kiosks/machines in the restaurant. The proposed system is expected to enable customers to pay using Touch 'n Go e-wallets. It will verify payment details through a payment gateway and generate a receipt upon receiving payment confirmation from the gateway. The receipt will include a food waiting number. Mr. Yong, an Information Systems Analyst, has been assigned to design the proposed self-service food ordering system.

Question 1

Based on the given scenario, which fact-finding technique is best suited for Mr. Yong to collect information from customers? Describe the selected fact-finding technique. (1+2 marks)

Questionnaires = 1 mark

Users are sent some questionnaires about the current system for them to answer at their leisure time. 2marks

Other acceptable answers: Interview, refer to lecture note : chapter 3

Question 2

Provide THREE (3) reasons to explain why this fact-finding technique is best suited for this scenario. (9 marks)

- **When there is a Large number of respondents.** Questionnaires technique is suitable to be used in a situation where there is a large number of restaurant customer to collect information.
- **Sensitive Issues.** In some circumstances the analyst may wish to collect information whose accuracy would be enhanced if the provider of the information remains anonymous
Useful in circumstances where the provider of the information wants to remain anonymous for sensitive issues.
- **Takes Time.** Where factual information requires the respondents to investigate, ^{further} ~~father~~ and compile data rather than respond with instant answers as in interviews. In this kind of situation, the respondents will be given sufficient time to carry their tasks.

Other acceptable answers : refer to lecture note : chapter 3

Question 3

Explain the following **TWO (2)** types of feasibility study.

(6 marks)

Technical feasibility (any 1 * 3 marks)

- Specifying the **hardware, software and IT personnel** performance requirements of the system under study. Evaluating the requirements whether the above requirements are **technically achievable**.
- This means that any proposed solution must be capable of being implemented using existing hardware, software and IT personnel in the organization.

Economic feasibility (3 marks)

- Identify the **costs and benefits** of implementing the system and defining when those benefits will be realized.

Other acceptable answer: refer to chapter 2

Question 4

Discuss possible feasibility issues related to the scenario.

(6 marks)

ONE (1) Technical feasibility issue :

Outputs (server)

- Assess the existing server to see if it can process food orders and generate food order receipts to customers on time.

ONE (1) Economic feasibility issue:

- Study on the selection of the most suitable technique for the investment eg Payback, etc.
- Assess the costs of purchasing new server and scanner (to scan QR code of customer tough n go E-wallet)
- Assess the costs of training for users of fast food restaurant.
- Determine and select the technical options under consideration the fast food restaurant should choose.

Other acceptable answers: refer to lecture note : chapter 2

Question 5

Draw a context diagram for the self-service food ordering system. (Note: ensure the Context Diagram shows only 1 system, 5 relevant data flows and 2 entities). (16 marks)

